

ECEN 662: Estimation & Detection Theory

Lecture 2: Minimal and Complete Sufficient Statistics

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- A sufficient statistic T is *minimal* if the partition induced by any other sufficient statistics is a *refinement* of the partition induced by T

A sufficient condition for minimality given sufficiency

- The minimality of a sufficient statistic can be verified based on the following proposition: A sufficient statistic T is *minimal* if for *any* $x, y \in \mathcal{X}$,

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} \text{ is independent of } \theta \quad \Rightarrow \quad T(x) = T(y)$$

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- Now suppose that $T'(x) = T'(y)$. We have

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \frac{g'_{\theta}(T'(x))h'(x)}{g'_{\theta}(T'(y))h'(y)} = \frac{h'(x)}{h'(y)}$$

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- By the above property, we must have $T(x) = T(y)$, which completes the proof of the minimality of T

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- Combined with the previous proposition, a statistic $T = T(X)$ is a *minimal sufficient* statistic if for *any* $x, y \in \mathcal{X}$,

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- The advantage of the above result is that it allows us to check sufficiency and minimality in one shot

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- Clearly, the ratio $f_{\theta}(x)/f_{\theta}(y)$ does *not* depend on θ *if and only if* $T(x) = T(y)$
- We thus conclude that for Bernoulli trials, $T = \sum_{i=1}^n X_i$ is a *minimal sufficient* statistic

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- (Proof) Let T be a *complete* sufficient statistic and T' be another sufficient statistic. We need to show that T is a function of T'
- Let

$$\begin{aligned}\psi(t') &:= \mathbb{E}_\theta[T | T' = t'] \\ \rho(t) &:= \mathbb{E}_\theta[\psi(T') | T = t]\end{aligned}$$

Note that both T' and T are *sufficient* statistics, so neither ψ nor ρ depends on θ

- Next, by the *law of total expectation*,

$$\begin{aligned}\mathbb{E}_\theta[T] &= \mathbb{E}_\theta[\mathbb{E}_\theta[T|T']] \\ &= \mathbb{E}_\theta[\psi(T')] \\ &= \mathbb{E}_\theta[\mathbb{E}_\theta[\psi(T')|T]] \\ &= \mathbb{E}_\theta[\rho(T)]\end{aligned}$$

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- Finally, recall the *law of total variance*:

$$\text{Var}[Y] = \mathbb{E}[\text{Var}[Y|Z]] + \text{Var}[\mathbb{E}[Y|Z]]$$

for any *real-valued* random variable Y

- For any $i \in [m]$, we have

$$\begin{aligned}\text{Var}_\theta[\rho_i(T)] &= \mathbb{E}_\theta[\text{Var}_\theta[\rho_i(T)|T']] + \text{Var}_\theta[\mathbb{E}_\theta[\rho_i(T)|T']] \\ &= \mathbb{E}_\theta[\text{Var}_\theta[\rho_i(T)|T']] + \text{Var}_\theta[\mathbb{E}_\theta[T_i|T']] \\ &= \mathbb{E}_\theta[\text{Var}_\theta[\rho_i(T)|T']] + \text{Var}_\theta[\psi_i(T')] \\ &= \mathbb{E}_\theta[\text{Var}_\theta[\rho_i(T)|T']] + \mathbb{E}_\theta[\text{Var}_\theta[\psi_i(T')|T]] + \text{Var}_\theta[\mathbb{E}_\theta[\psi_i(T')|T]] \\ &= \mathbb{E}_\theta[\text{Var}_\theta[\rho_i(T)|T']] + \mathbb{E}_\theta[\text{Var}_\theta[\psi_i(T')|T]] + \text{Var}_\theta[\rho_i(T)]\end{aligned}$$

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\end{aligned}$$

- Since variances are always *non-negative*, we must have

$$\begin{aligned}
\text{Var}_\theta[\rho_i(T)|T'] &= 0 && \text{with probability 1} \\
\text{Var}_\theta[\psi_i(T')|T] &= 0 && \text{with probability 1}
\end{aligned}$$

for all $i \in [m]$. That is, $\rho(T)$ must be a *deterministic* function of T'

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- The “mild conditions” under which the result is valid are that the various expectations and variances used throughout the proof are well defined and finite

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$$\begin{aligned}\mathbb{E}_{\theta}[\phi(T)] &= \sum_{t=0}^n \phi(t) \binom{n}{t} \theta^t (1-\theta)^{n-t} \\ &= (1-\theta)^n \sum_{t=0}^n \phi(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t \\ &= 0\end{aligned}$$

we must have

$$\sum_{t=0}^n \phi(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t = 0$$

for all $\theta \in (0, 1)$

- Let $\eta = \frac{\theta}{1-\theta}$. We have

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- If for some real-valued function ϕ we have

$$\mathbb{E}_{\theta}[\phi(T)] = n\theta^{-n} \int_0^{\theta} \phi(t)t^{n-1}dt = 0$$

for all $\theta > 0$, we must have

$$\int_0^{\theta} \phi(t)t^{n-1}dt = 0, \quad \forall \theta > 0$$

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$$\begin{aligned} f_{\theta}(x) &= \prod_{i=1}^n \left(\frac{1}{\sqrt{2\pi\theta^2}} \exp \left(-\frac{(x_i - \theta)^2}{2\theta^2} \right) \right) \\ &= \left(\frac{1}{\sqrt{2\pi\theta^2}} \right)^n \exp \left(-\frac{\sum_{i=1}^n x_i^2}{2\theta^2} + \frac{\sum_{i=1}^n x_i}{\theta} - \frac{n}{2} \right) \end{aligned}$$

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- By the Fisher-Neyman factorization theorem,

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is a two-dimensional sufficient statistic for θ

- The likelihood ratio

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \exp \left(-\frac{\sum_{i=1}^n x_i^2 - \sum_{i=1}^n y_i^2}{2\theta^2} + \frac{\sum_{i=1}^n x_i - \sum_{i=1}^n y_i}{\theta} \right)$$

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- Therefore, the likelihood ratio $f_{\theta}(x)/f_{\theta}(y)$ is constant over $\theta > 0$ only if

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- We may thus conclude that

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is in fact a *minimal* sufficient statistic for θ

- In your homework assignment, you are asked to find a real-valued function $\phi(t)$ such that $\mathbb{E}_\theta[\phi(T)] = 0$ for all $\theta > 0$ but $\mathbb{P}_\theta(\phi(T) = 0) < 1$ for some $\theta > 0$, i.e., T is *not* complete

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- Consequently, if there is a minimal sufficient statistic that is *not* complete, then a complete sufficient does *not* exist

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- Therefore, in general, a minimal sufficient statistic is *not* necessarily complete
- In fact, if a complete sufficient statistic exists, then any minimal sufficient statistic is also complete because it is a function of the complete sufficient statistic
- Consequently, if there is a minimal sufficient statistic that is *not* complete, then a complete sufficient does *not* exist
- As we will see in later lectures, while both minimal and complete sufficient statistics are “optimal” in terms of data reduction, only complete sufficient statistics guarantee optimality when it comes to *risk reduction*

Next lecture

Exponential family of distributions